# Somtochukwu David Ogbonna

COMPUTATIONAL BIOLOGY · COMPUTER SCIENCE
BIOINFORMATICS · MACHINE LEARNING · SOFTWARE ENGINEERING

— BIOINFORMATICS | 28 YEARS OLD | REMOTE-READY

## PROFILE

**Computational biologist and software engineer** working at the intersection of biology and computer science. Holds a master's degree in bioinformatics and a bachelor's degree in computer science with a biology minor, both completed in Canada. Builds pipelines that turn raw sequencing and laboratory data into analysis-ready datasets, writes the software that runs them, and applies machine learning where the biology is too complex for a fixed rule. Comfortable moving between a genome alignment problem, a Python service, a statistical model, and a conversation with wet-lab scientists. Based in Toronto with a track record of remote and hybrid delivery for research groups, biotech teams, and health-data organisations across North America.

## TWO DOMAINS, ONE PRACTICE

LIFE SCIENCES

## Computational Biology & Bioinformatics

Trained in molecular biology and genetics with applied work in genomics, transcriptomics, and laboratory data analysis. Fluent in the file formats, reference databases, and statistical assumptions that genomics workflows depend on.

- Genomic and transcriptomic sequence analysis
- Variant calling, annotation, and filtering
- RNA-seq quantification and differential expression
- Single-cell data processing and clustering
- Phylogenetics and comparative genomics
- Experimental design and statistical power analysis

COMPUTING

## Computer Science & Software Engineering

Strong software fundamentals applied to scientific computing — not just notebooks, but tested, versioned, reproducible code that other researchers and engineers can run, extend, and trust.

- Python and R for scientific computing
- Algorithm design and complexity analysis
- Workflow orchestration and pipeline engineering
- Data modelling, SQL, and database design
- Machine learning and statistical modelling
- Cloud infrastructure and containerised deployment

## PROFESSIONAL EXPERIENCE

### Computational Biologist & Software Engineer

2023 — PRESENT

Genome Systems Lab · Toronto, Ontario REMOTE

- Design and maintain the primary analysis pipeline for a cancer genomics research programme, processing several hundred whole-exome and RNA-seq samples per year from raw reads through to annotated variant calls.
- Rewrote the variant-calling workflow as a containerised Nextflow pipeline with automated quality gates, cutting per-sample runtime by roughly forty percent and removing a recurring class of manual reruns.
- Built a Python service that exposes the lab's reference datasets and analysis results through a documented API, now used by three collaborating research groups.
- Developed a machine learning classifier for distinguishing likely pathogenic from benign variants, combining sequence features, population frequency, and predicted structural impact; validated against a held-out clinical set.
- Introduced version control, automated testing, and reproducible environment management to a group that previously relied on ad-hoc scripts, reducing analysis discrepancies between team members to near zero.
- Translate requirements directly with wet-lab scientists and clinical collaborators, running regular working sessions to keep analysis aligned with experimental design.

PYTHON NEXTFLOW DOCKER GATK SCIKIT-LEARN AWS

### Bioinformatics Analyst

2022 — 2023

Meridian Health Data · Mississauga, Ontario HYBRID

- Processed and quality-controlled RNA-seq datasets for a multi-site biomarker study, building the quantification and differential expression workflow in R and Python.
- Developed reproducible R Markdown reporting templates that replaced manually assembled slide decks, cutting weekly reporting effort substantially.
- Built a single-cell analysis workflow for clustering and cell-type annotation across several thousand cells, including batch correction and marker validation.
- Designed the study database schema and data dictionary, improving traceability from raw sample to published figure.
- Presented analysis results to clinical and non-technical stakeholders, translating statistical output into plain-language findings.

R SEURAT DESEQ2 POSTGRESQL JUPYTER

### Research Software Developer

2021 — 2022

University of Toronto Research Computing · Toronto, Ontario REMOTE

- Contributed to an open-source toolkit for comparative genomics, implementing alignment parsing and distance calculation modules in Python and C++.
- Profiled and optimised core routines, reducing memory footprint on large genome comparisons and enabling runs that previously exceeded available resources.
- Wrote unit and integration tests for the parsing layer and set up continuous integration, raising coverage on the components under active development.
- Authored installation and usage documentation adopted by external research groups using the toolkit.

PYTHON · C++ · GIT · CI/CD · LINUX

### Teaching Assistant — Computational Biology

2020 – 2021

Graduate Department of Biology · Toronto, Ontario

- Led laboratory and tutorial sessions for an undergraduate computational biology course covering sequence analysis, alignment, and phylogenetic methods.
- Developed worked examples and starter code that reduced student setup friction on bioinformatics assignments.
- Held weekly office hours and graded assignments, providing written feedback on both biological reasoning and code quality.

## TECHNICAL SKILLS

### PROGRAMMING

| | |
|---|---|
| Python | EXPERT |
| R | ADVANCED |
| SQL | ADVANCED |
| C++ | WORKING |
| Bash / Shell | ADVANCED |

### BIOINFORMATICS

| | |
|---|---|
| GATK · BWA · samtools | ADVANCED |
| DESeq2 · edgeR | ADVANCED |
| Seurat · Scanpy | WORKING |
| Nextflow · Snakemake | ADVANCED |
| Bioconductor · Biopython | ADVANCED |

### DATA & ML

| | |
|---|---|
| pandas · NumPy | EXPERT |
| scikit-learn | ADVANCED |
| PyTorch | WORKING |
| Statistical modelling | ADVANCED |
| Data visualization | ADVANCED |

## TOOLS & PLATFORMS

Git · GitHub · GitLab | Docker | Kubernetes | AWS (S3, EC2, Batch)
Linux / HPC clusters | PostgreSQL | MongoDB | Jupyter | RStudio | Jira
Confluence | Slack | Zoom | QIIME 2 | IGV | UCSC Genome Browser
Ensembl / NCBI resources

## EDUCATION

### M.Sc. in Bioinformatics

2021 – 2023

University of Toronto · Toronto, Ontario, Canada

**Thesis:** "Machine Learning Approaches to Variant Pathogenicity Prediction Using Sequence and Structural Features."
**Focus areas:** genomics, statistical genetics, machine learning, algorithm design, high-performance computing for biological data.

### B.Sc. (Hons) in Computer Science

2017 – 2021

York University · Toronto, Ontario, Canada

**Minor:** Biology
**Core coursework:** Data Structures and Algorithms, Operating Systems, Databases, Software Engineering, Machine Learning, Molecular Biology, Genetics.

### Certificate in Data Science

2022 – 2023

Continuing Studies · Toronto, Ontario, Canada

**Coverage:** statistical inference, predictive modelling, data engineering, reproducible research practice, and applied machine learning workflows.

## RESEARCH & PRESENTATIONS

- S. D. Ogbonna, “Machine Learning Approaches to Variant Pathogenicity Prediction Using Sequence and Structural Features,” *M.Sc. thesis, University of Toronto*, 2023.
- S. D. Ogbonna and R. A. Whitfield, “Feature Engineering Strategies for Rare Variant Classification in Exome Cohorts,” *presented at the Canadian Bioinformatics Workshop*, 2022.
- Contributor, open-source comparative genomics toolkit — alignment parsing and distance calculation modules, *public repository*, 2021–2022.
- S. D. Ogbonna, “Reproducible Pipelines for Small Research Groups,” *internal seminar, Genome Systems Lab*, 2024.

## LANGUAGES

English — native / professional C2

French — reading knowledge A2

Igbo — heritage B1

Scientific/technical writing ADVANCED

## AVAILABILITY

**WORKING SETUP**

Based in Toronto, Ontario, with a fully equipped home office and reliable high-speed connectivity. Experienced in remote and hybrid delivery for research groups, biotech teams, and health-data organisations across Canadian and United States time zones, including asynchronous collaboration, structured handovers, and reproducible analysis documentation. **Available for full-time roles, contract engagements, and project-based bioinformatics or scientific software work.** Legally authorised to work in Canada.